

RAMCO INSTITUTE OF TECHNOLOGY
RAJAPALAYAM

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DEPT. OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Year and Semester: II/IV

Subject code and Title: CS3591-COMPUTER NETWORKS

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Active learning:

Demonstration on Physical Network Components using Active Learning Method

Objective

To engage learners in an interactive demonstration that explores the functions and interconnectivity of key physical network components such as **Switches, Routers, Access Points**, and **Point-to-Point Network Connections**, enabling them to visualize real-world networking scenarios and apply theoretical knowledge practically.

Network components used:

- Network Switch (Layer 2)
- Router (Layer 3)
- Wireless Access Point
- Ethernet cables
- Network Interface Cards (NICs)
- Laptops or desktop PCs
- Two wireless-enabled devices (for Access Point)
- Two networking nodes (for Point-to-Point setup)
- Network simulation software (optional – e.g., Cisco Packet Tracer)

Procedure

Step 1: Introduction (5-10 minutes)

- Briefly explain the objective of the session.
- Introduce the physical components to be demonstrated.
- Emphasize the importance of understanding network hardware in real environments.

Step 2: Switch Demonstration

- Show a physical switch.
- Connect multiple computers using Ethernet cables to the switch.
- Demonstrate communication between devices using ping or file sharing.

- Explain how switches operate on Layer 2 (Data Link Layer) and use MAC addresses for forwarding.

Key Points:

- Switching vs. Hub
- MAC Address Table (CAM Table)
- Collision domains

Step 3: Router Demonstration

- Introduce the **router** and explain Layer 3 (Network Layer) functionality.
- Connect two separate networks (e.g., two switches) using the router.
- Configure IP addresses on the connected devices.
- Demonstrate inter-network communication.

Key Points:

- Routing between different subnets
- Default Gateway concept
- Basic static routing setup

Step 4: Wireless Access Point (AP)

- Connect and power on a wireless access point.
- Connect wireless devices (smartphones or laptops) to the AP.
- Show internet access or file transfer over the Wi-Fi connection.
- Discuss SSID, frequency bands (2.4 GHz / 5 GHz), and wireless standards (802.11n/ac/ax).

Key Points:

- Wireless coverage area
- Encryption methods (WPA2/WPA3)
- Differences between AP and router

Step 5: Point-to-Point Network Setup

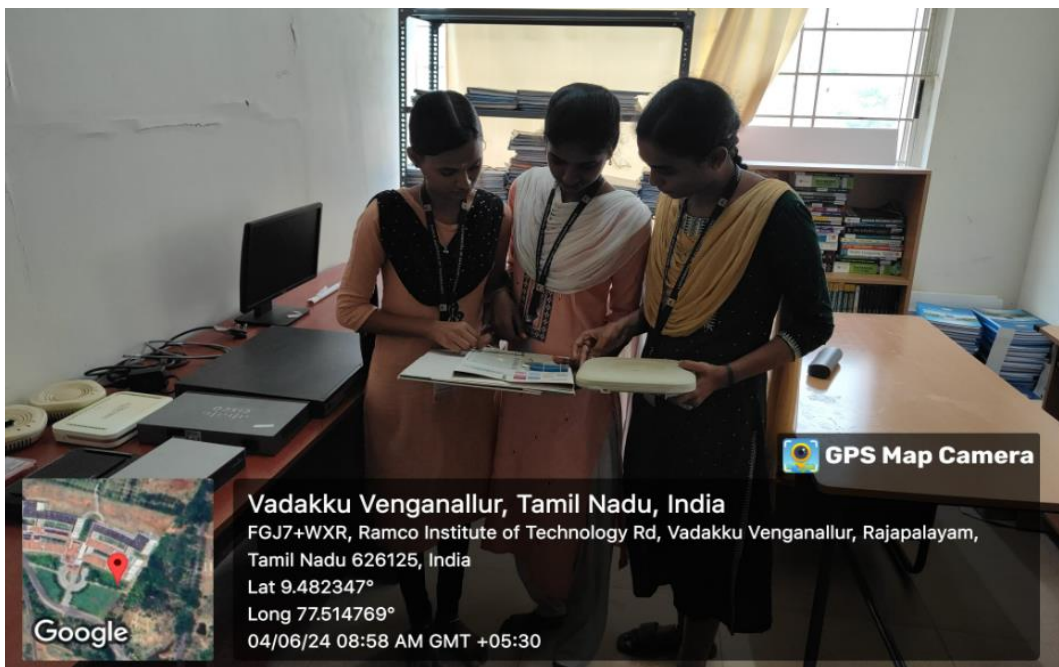
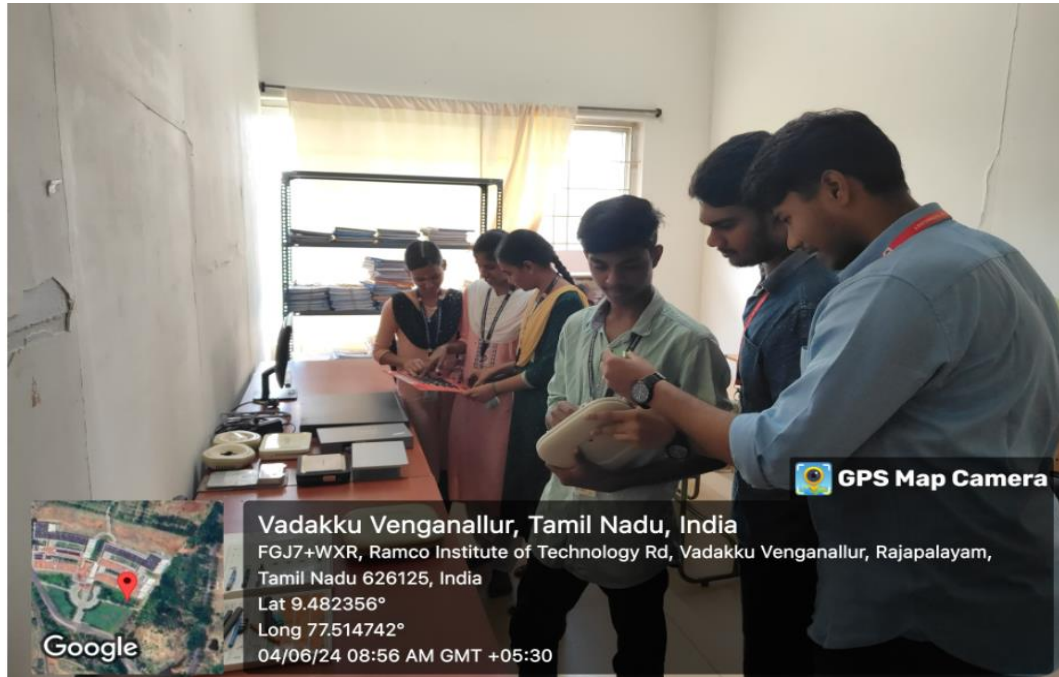
- Set up a direct wired or wireless connection between two nodes (e.g., two laptops or routers).
- Assign IP addresses manually.
- Demonstrate a peer-to-peer communication model (file sharing, ping, etc.).
- Use directional antennas if demonstrating wireless point-to-point.

Key Points:

- Suitable for long-distance networking
- Use in CCTV, inter-building communication
- Line-of-sight requirements in wireless P2P

Step 6: Learner Interaction

- Ask students to replicate small setups (e.g., plug in cables, assign IPs).
- Encourage questions and problem-solving in real-time.
- Use guided inquiry (e.g., "What happens if we remove the router?")



Step 7: Summary and Recap

- Recap the functionality of each component.
- Use a whiteboard to draw the final complete network topology.
- Link practical observations to theoretical networking models (OSI/ TCP-IP).

Reflective Critique:**Effectiveness of Method:**

The demonstration-based active learning method was highly effective in bridging the gap between theory and practice. Learners engaged directly with hardware components, which enhanced their conceptual clarity and confidence in handling real-world networking setups.

What Worked Well:

- Hands-on exposure made abstract concepts tangible.
- Immediate visual feedback from successful connections (e.g., ping replies) reinforced learning.
- Peer collaboration encouraged active participation and dialogue.

Challenges Faced:

- Time management was a challenge; each component required more explanation than expected.
- Some students lacked foundational knowledge (e.g., IP addressing), slowing down progress.
- Technical issues like loose cables or misconfigured settings created brief delays but became learning moments.

Learner Response:

Most students were enthusiastic and engaged, expressing interest in more lab-based sessions. Those with prior theoretical knowledge grasped the content quicker, while others needed extra support, suggesting future sessions might benefit from tiered instruction or pre-session reading.

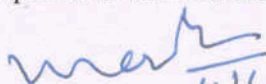
Improvements for Next Time:

- Provide pre-session materials on IP addressing, OSI model, etc.
- Assign students into small teams with rotating roles (e.g., configurator, tester, observer).
- Use network simulators in parallel to reinforce learning when hardware is limited.
- Include a checklist-based worksheet to guide learners during the demo.

Conclusion:

The demonstration of physical network components using an active learning approach proved to be an engaging and educational experience. It highlighted the importance of practical exposure in networking education and revealed areas where scaffolding and support can improve learner outcomes. Future sessions will be adapted to include more learner-led exploration and scenario-based problem-solving.

Faculty Incharge


4/6/24

HOD/AD


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